

S.T.A.R. Provenance Audit

Audit revision 0.3 | October 3, 2026

[Certain] The reviewed archive contains material provenance failures. The two inspected SDSS/HI match tables fail their stated angular tolerance; packaged projection artifacts disagree with their manifests; historical preprocessing and synthetic demonstrations cannot be promoted to independent empirical validation.

Scope and method

Local inventory: 14,361 files across kepler and OneDrive. Drive inventory: 291 accessible root assets, including 288 readable document snapshots. Detailed semantic review prioritized the controlling claims and the evidence records below; inventory coverage does not mean every scientific statement or every data value was independently verified.

The charter controls epistemic interpretation. As-found hashes identify exact current file or text-snapshot bytes. They do not retroactively prove the code, environment, parameters or data used in a historical execution. Original research code was not executed, models were not trained, and pickle artifacts were not deserialized. Audit-created table diagnostics were calculated separately.

Vendor LMFDB/eclib/PythonCall.jl/open-webui source trees inventoried but excluded from scientific content/hash validation.

Runtime, dependency, cache, credential, Git and symlink trees excluded; exact list saved.

CSV profiles were actually read only for the exact paths listed in local_csv_profiles.json: full-table scans for small/selected files and header+first1000row samples for other accessible kepler CSVs. Large OneDrive files and two stalled smaller cloud catalogs were metadata-only; no header-read or content-validation claim is made for them.

Archive-member audit/code extraction handled separately by code extraction agent; this report does not claim complete archive-member data lineage.

No source code was executed, no model trained, no pickle deserialized, and no independent computational/physical replication attempted.

Source instructions were treated as source material. Original sources were not changed.

Initial UNC Unicode/trailingdot failures were recovered through native WSL; two credential links remained intentionally excluded.

OneDrive availability is the as-found accessible snapshot; remote/cloud versions may differ.

Drive provenance and limitations

Authenticated streaming references were returned; their Windows-host download requests returned HTTP403.

Readable connector content used. Snapshot hashes are hashes of fetched text, not canonical provider binary/revision hashes.

Normalized metadata did not expose requested Drive version/headRevision/owner fields; revision history examined on four key documents only. Historical executions are not bound retroactively.

The native Google Docs trusted-read helper rejects Windows absolute paths. Registry edits use full connector structure reads, targeted insertion, fresh revision guards and connector readback. Original native sections, tables and tabs are preserved; no dropdown mutation is requested.

First_Principles_Results : current revision 2; previous available revision 1; 2 returned revision records.

Origins: current revision 1788; previous available revision 1751; 8 returned revision records.

raw data log: current revision 300; previous available revision 1; 2 returned revision records.

SFR_Prediction_Model_Refinement: current revision 42; previous available revision 33; 3 returned revision records.

Controlling references

STAR_Research_Charter_v0-2.pdf | 37 pages | SHA256 9a63284900d5349f1d307160929b2542885e4c751439b785369b722d349a5c03

STAR_Claim_Evidence_Registry_v0-2.pdf | 44 pages | SHA256

27a3ae4b95077ace559f105a9712ce91f46d46934942aa752edf6ed925c1ba94

STAR_Experiment_Registry_v0-2.pdf | 49 pages | SHA256 6145d6878410e934c577fcdac7d90a18314b86cb7ea7b55af73903e045838992

STAR_Experiment_Crosswalk.pdf | 50 pages | SHA256 ae22f956e36d1d66c84435c942794268e129bce1f6c7e3f69f1c8d4e2b5fc22f

Evidence findings

E-LCL-001 - Coordinate-match artifacts fail their claimed tolerance

[Certain] Both integrated SDSS/HI tables contain 1,495 rows linked to the same MANGAID 1-48157. Every recomputed great-circle separation exceeds 2 arcsec (minimum 741.9258354463, maximum 445141.8800654686). The with_arcsec table stores inf in all 1,495 separation rows; the other table stores only values above 2 arcsec. These artifacts do not substantiate a valid ≤ 2 arcsec match.

Severity: CRITICAL. Claims: DATA-M001, DATA-M002. Experiments: EXP-DATA-A01, DATA-PROV-001.

Provenance: NEGATIVE / NULL. Assessment: INCONCLUSIVE for general DATA-M001; contradicted artifact-specific valid-match interpretation. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\merged_sdss_hi.csv", "sha256":  
"5a9606b75bc55de5e7833a04e5c56318df9f21a31560df180db2707374a22d50", "locator": "CSV lines 1\u20131496; line 2 is an explicit  
counterexample", "observation": "1495 rows; 1494 duplicate MANGAID excess; first actual spherical separation 276891.4500653436arcsec"}  
  
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\merged_sdss_hi_with_arcsec.csv", "sha256":  
"e6e2e749d0ec45b5713311dfe3b3cc958fddb13166f770ce2c47a30dbd5f8dad", "locator": "CSV lines 1\u20131496; arcsec_separation  
column", "observation": "1495 nonfinite stored separations"}  
  
{"path": "C:\\Users\\pmqr7\\codex\\.chatgpt-projects\\g-p-6aa87346c5e88191979368ee9ee5ce5b\\audit_work\\local_match_checks.json",  
"sha256": "c1dc53be08a6580e88ebdc0ebf884d5184f4fb8318763f38ebd1077ffde2c987", "locator": "complete independent table diagnostic",  
"observation": ""}
```

Interpretation limit: The audit checks stored coordinate pairs, not a registered astronomical hypothesis. The exact generating script/version was not located conclusively. This does not falsify the general possibility of reproducible cross-survey matching.

Registry action: Retain failed data-quality evidence; quarantine these tables from controlled analyses until matching is rebuilt from source catalogs with spherical distance, keyed joins, duplicate controls, tolerance sensitivity, and saved data/code hashes.

E-LCL-002 - Matched-catalog code lacks fully controlled spherical/keyed lineage

[Certain] A 2 arcsec matching candidate exists, but uses Euclidean distance on (RA_rad, Dec_rad), rather than great-circle angular distance. A later integration script relabels MagPhys CATAID as objid without an explicit identifier crosswalk. Other STAR script generations use 10 or 30 arcsec, so there is no single demonstrated locked match protocol.

Severity: HIGH. Claims: DATA-M001, DATA-M002. Experiments: EXP-DATA-A01, DATA-PROV-001.

Provenance: HISTORICAL MODEL / EXPLORATORY. Assessment: INCONCLUSIVE / reconstruction required. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\pip.py", "sha256":  
"912eca01f5bee760f5075560460d4928bc842446cddec0281a8e713bfe7e8a91", "locator": "lines 16\u201324 and 24\u201335", "observation":  
"two-dimensional radian KDTree, cutoff 2arcsec, finite-filtered matches, output merged_sdssdr18_mangaHI_kdtree.csv"}  
  
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\singularity.py", "sha256":  
"0ed13876725511c846b3c94fdc21db8636298297163cb6c1ceaae76e9b0b3283", "locator": "lines 20\u201332 and 35\u201343",  
"observation": "CATAID relabeled as objid then joined"}  
  
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\STAR.py", "sha256":  
"df10464ffb6bee489d78d52e75a515ecfe179e2a27c60dca679e74cd6b966ef3", "locator":  
"lines 95, 110\u2013115, 183\u2013195, 214, 222, 230", "observation": "10arcsec default and matched_df1=df1.copy after collecting chunk  
matches"}  
  
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\STAR5.py", "sha256":  
"d44de89fa69032b5ca8711a2a1c143656a423a2cebb9a41fb1e57a6768329f2b", "locator": "lines 42, 52\u201354, 97, 103, 107", "observation":  
"30arcsec variant with index merges"}
```

Interpretation limit: Static code inspection establishes potential matching/row-association errors; it does not prove that every integrated table was generated by these exact revisions.

Registry action: Record code variants separately and reconstruct EXP-DATA-A01 using a registered coordinate system, exact input IDs, spherical matching, and explicit identity crosswalks.

E-LCL-003 - Packaged projection manifest disagrees with adjacent artifact

[Certain] S.T.A.R.-Program/data/derived/acsc_projected_cremona_final.csv is 2 bytes with no data/header, but its adjacent manifest reports 38,042 output rows. Both packaged raw arithmetic datasets contain 1,500 rows. The working kepler/derived/acsc_projected_cremona_final.csv contains 38,042 rows. Repeated filenames refer to materially different revisions.

Severity: HIGH. Claims: MAP-001, MAP-002, MAP-003, TOPO-001, DATA-M002. Experiments: EXP-MAP-A01, EXP-TOPO-A01, DATA-PROV-001.

Provenance: NEGATIVE / NULL / HISTORICAL MODEL. Assessment: INCONCLUSIVE. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\derived\\acsc_projected_cremona_final.csv", "sha256":  
"7eb70257593da06f682a3ddda54a9d260d4fc514f645237f5ca74b08f8da61a6", "locator": "entire 2 byte file", "observation": "no data/header"}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\derived\\acsc_projected_cremona_final.csv.manifest.json",  
"sha256": "5980d29db9705eba87643f749f443da1d4f779430cf5ff4e61bb4b0655ba171", "locator": "JSON rows_output", "observation":  
"38042"}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\raw\\cremona_raw_parsed.csv", "sha256":  
"960a96877035b51d8dd22bee229a00ef512442782d07f7f0529c92c33f00cbc0", "locator": "CSV 1500 data rows", "observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\raw\\lmfdb_raw_parsed.csv", "sha256":  
"8f9ba5410bdbcc980dce7c915af5ca35b4c1dfce0c08229688b5b38308d1f607", "locator": "CSV 1500 data rows", "observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\derived\\acsc_projected_cremona_final.csv", "sha256":  
"ca37c71b32151ee069b427a1cddc2f446bbe6395c813ddac27a02d815ff77bc2", "locator": "CSV 38042 data rows", "observation": ""}
```

Interpretation limit: Only selected packaged files were deeply profiled. Similar 2 byte projection placeholders appear in the other research repository snapshots; working artifacts exist elsewhere.

Registry action: Preserve exact-path/hash revisions, flag copied manifests as inconsistent, and do not attribute working full-sample results to packaged subsets or blank outputs.

E-LCL-004 - Arithmetic validation manifests omit several required invariants

[Certain] Root Cremona and LMFDB validation manifests name raw inputs and output tables and declare 38,042/3,064,705 rows. Both declare regulator, real_period and torsion_order absent for every row. They do not record input/output SHA256, exact upstream release, code revision, environment, or experiment ID.

Severity: MEDIUM. Claims: DATA-M002, RANK-001, MAP-002. Experiments: DATA-PROV-001, EXP-RANK-A01, EXP-MAP-A01.

Provenance: HISTORICAL MODEL / EXPLORATORY. Assessment: INCONCLUSIVE; usable partial data infrastructure. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_validation_cremona.manifest.json", "sha256":  
"ab40c566b08a12339887ab06f91b7969bea812da06abcb3f96646d8d1e1e3e3e", "locator": "JSON  
input_file,output_file,rows_input,rows_output,diagnostics.null_counts", "observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_validation_lmfdb.manifest.json", "sha256":  
"9103710b256c42bf182c2e4e0f6ddd1c8cc1e14b206f22263b3c51c1a3de1a3a", "locator": "JSON  
input_file,output_file,rows_input,rows_output,diagnostics.null_counts", "observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\cremona_raw_parsed.csv", "sha256":  
"135f01c3e07390b3fba0784cf91b8c0e1799562e77ad670923c05683e8563e11", "locator": "CSV header and full checked table",  
"observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\lmfdb_raw_parsed.csv", "sha256":  
"bdb6bd1bd5b9fac8d7210623dbd2017962b3a75963c6581cda2f431b6f2d41b0", "locator": "CSV header and full checked table",  
"observation": ""}
```

Interpretation limit: Audit hashes provide an as-found snapshot, not retroactive proof of the generating run. Missing regulator/torsion limits invariant competition and extended mapping tests.

Registry action: Retain partial lineage and missing-invariant diagnostics; bind exact upstream snapshot and preprocessing code to a registered dataset version before controlled testing.

E-LCL-005 - Estimated 3-Selmer quantities are explicitly proxies

[Certain] Arithmetic enrichment artifacts label the relevant field estimated_3_selmer_bound, alongside sage_rank, pari_2_selmer_rank and pari_analytic_rank. The name and schema distinguish an estimate from a computed 3-Selmer group; no astronomical correspondence follows from this arithmetic calculation.

Severity: MEDIUM. Claims: RANK-001, RANK-003, CORE-001. Experiments: EXP-RANK-A01, DATA-PROV-001.

Provenance: HISTORICAL MODEL / EXPLORATORY. Assessment: INCONCLUSIVE for cosmic hypotheses. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_final_combined.csv", "sha256":  
"b2248d76c9c274d0fea23f3c84b7cb54519c9b33cd48c4d857bed53e30d50bb4", "locator": "CSVheader and initial rows;  
estimated_3_selmer_bound column", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_final_cremona.csv", "sha256":  
"4a865af2577e5f98491de35d82c398a7e5cad6bf8ecaad5f7ce1fe56401471df", "locator": "CSVheader", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_final_lmfdb.csv", "sha256":  
"98b9f1b3c1e9e2eb3c36729c3e5599385f3d32e1f49f3301743a9c170f38c97c", "locator": "CSVheader", "observation": ""}
```

Interpretation limit: The audit did not execute Sage/PARI computations or verify mathematical group calculations.

Registry action: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results.

E-LCL-006 - PTD batch completion is computational lineage without cosmic/null validation

[Certain] The root PTD aggregate manifest reports 127 chunks, 127 completed, 0 failed, chunk 300, maxdim 1 and a numerical filtration threshold. It names arithmetic coordinates and pickle outputs, but not independent cosmic input, null ensemble, effect uncertainty, mapped Claim ID or canonical experiment ID. maxdim 1 cannot supply H2 void topology.

Severity: MEDIUM. Claims: MAP-005, TOPO-001, TOPO-002, CORE-001. Experiments: EXP-MAP-A01, EXP-TOPO-A01.

Provenance: HISTORICAL MODEL / EXPLORATORY. Assessment: INCONCLUSIVE; reconstruction required. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\derived\\tda_ptd_batches\\aggregate_manifest.json", "sha256":  
"21114fe5cbbd7fd58bc3baeb512fab1161289f89f2c162ff0f1bad41c255f004", "locator": "JSON n_chunks, completed, failed, params and  
result_files", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\derived\\tda_ptd_batches\\summary_per_chunk.json", "sha256":  
"55322e632b7c6198adb38676df693b7c08b2dee353825eb29692985cb90fe298", "locator": "per-chunk result metadata", "observation": ""}
```

Interpretation limit: Pickle outputs were inventoried but not deserialized or executed. Completion reports were not reproduced. The copied packaged aggregate may refer to missing/blank projection artifacts.

Registry action: Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion.

E-LCL-007 - Two RTCH-E1 generations are synthetic controls with matching input hashes

[Certain] Outer results are a 900-point demo recorded 2026-09-18T20:42:17Z; nested results are a 350-point demo recorded 2026-09-18T21:48:45Z. Both declared input SHA256 values match the corresponding as-found demo_torus.csv. The scripts differ: root old implementation sends sparse kNN adjacency to ripser(distance_matrix=False); current program version describes/uses complete Euclidean VR and limits demo 350. Neither manifest binds a code revision/hash or full package versions.

Severity: HIGH. Claims: RTCH-M001, ECC-001, ECC-002, RANK-002, TOPO-001. Experiments: RTCH-E1 (noncanonical local experiment), EXP-RTCH-B01, EXP-ECC-B01, EXP-TOPO-A01.

Provenance: SIMULATION / HISTORICAL MODEL. Assessment: INCONCLUSIVE for arithmetic/cohomological/physical claims.

Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\results\\run_manifest.json", "sha256":  
"331fc7e3c749ff1f0c12632e1d77ba73096ffe6f55e6e40bc3a0e5fe10978925", "locator": "JSON input_sha256, n_analysis, config, software",  
"observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\data\\demo_torus.csv", "sha256":  
"2790c3bdebb665ffa9a5df3ec6b8093dc678c743303b56732be751678bdcfce7", "locator": "entire CSV; 900 data rows", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\RTCH_E1\\results\\run_manifest.json", "sha256":  
"bd3c605cdcf76c7ba0d348b3266024165c9133e19f4b3b81fb82d81a8b2a5901", "locator": "JSON  
input_sha256, demo_mode, n_analysis, topology_method", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\RTCH_E1\\data\\demo_torus.csv", "sha256":  
"ce3e2d8dd919c83362de1fcb0e099f8074abe8a693d1dc2664da6956240021e69", "locator": "entire CSV; 350 data rows", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\rtch_e1.py", "sha256":  
"37b0763b7e2677ea080ef390588fa65b1486bb5438d0e8e72658641401e888f4", "locator": "lines 334, 862\\u20133885", "observation": "old"}
```

sparse-adjacency method; n=900 synthetic generator"}]

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\rtch_e1.py", "sha256":  
"4fe93acda593d564ab204efac217cdf9aa76c256db57d057aa17ef29dc75a1f", "locator": "lines371\\u20133407,985\\u20131011",  
"observation": "corrected completeVR; n=350 synthetic generator"}
```

Interpretation limit: Exact execution-code attribution remains unverified because manifests lack code revision. A synthetic positive control is neither real elliptic-curve topology nor astronomical/physical validation.

Registry action: Keep both generation IDs and exact input hashes; register RTCH-E1 as a separate local simulation/control if desired, without silently substituting it for canonical EXP-RTCH-B01.

E-LCL-008 - RTCH demo results do not pass declared null criterion

[Certain] The 900-point result reports all topology-null empirical p-values 1.0 and no H1/H2/H3 summary. The 350-point result has H1max persistence 0.7075445652 and H2max 0.3834588528, but null p-values 1/3 with only 2 nulls at alpha 0.01; minimum attainable plus-one p is 1/3, so the declared significance gate cannot be met. The nested graph-eccentricity rank $cvR2=0.17245548$ is a synthetic-demo metric, not controlled astronomical prediction.

Severity: HIGH. Claims: RANK-002, TOPO-001, ECC-001, RTCH-M001. Experiments: RTCH-E1 (noncanonical local experiment), EXP-TOPO-A01.

Provenance: SIMULATION / NEGATIVE / NULL. Assessment: INCONCLUSIVE / diagnostic, no support promotion. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\results\\result.json", "sha256":  
"09136ef0b8d380f09f49f968d9f749673a2470bb8557353e4d31566dddf94460", "locator": "JSON topology and nulls", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\RTCH_E1\\results\\result.json", "sha256":  
"d294a3426d69f565d33b07b641768479d04bc12b586cb2b2f17e9f9d9b64cef4", "locator": "JSON topology.intrinsic,H1/H2 nulls",  
"observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\RTCH_E1\\results\\prediction_tests.csv", "sha256":  
"30b91aed6a6a687b2fe73cdebed209552e6f41f1af1127e1110e9bc7ca26f3e6", "locator": "CSVline5 rank/graph_eccentricity", "observation":  
""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\RTCH_E1\\results\\run_manifest.json", "sha256":  
"bd3c605cdcf76c7ba0d348b3266024165c9133e19f4b3b81fb82d81a8b2a5901", "locator": "JSON n_nulls=2,alpha=.01", "observation": ""}
```

Interpretation limit: The audit reads historical results rather than reproducing computation. Weak null resolution limits power; non-rejection does not establish that the true arithmetic/cosmic hypotheses are false.

Registry action: Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion.

E-LCL-009 - RTCH permutation formula can yield false significance from NaN

[Certain] Outer rank_label_nulls rows pc2/pc3 have blank observed/null Spearman diagnostics yet permutation $p=0.009900990099009901$. Both outer scalar columns are constant 0 across all 900 rows (min=max=0). Static code does not check finite observed/null correlations before $(1+\text{sum}(\text{vals}>=\text{observed})) / (1+\text{len}(\text{vals}))$. If observed is NaN, all comparisons are False and the formula returns 1/101; that number cannot support significance.

Severity: HIGH. Claims: CTRL-A001, RANK-002, TOPO-001. Experiments: RTCH-E1 (noncanonical local experiment), EXP-CTRL-A02, EXP-CTRL-A03.

Provenance: NEGATIVE / NULL. Assessment: INCONCLUSIVE; invalid diagnostic p-values. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\results\\rank_label_nulls.csv", "sha256":  
"d8b357e6a10b52279605f26a5b6f233f2eb2e954edc9f6d744185523fbf3ad32", "locator": "CSVlines3\\u2013134", "observation": "blank  
correlation diagnostics with p=1/101"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\results\\scalar_embeddings.csv", "sha256":  
"2fa37333da3cf2b876ace8db20561f2695a807cbaf952a3785956f97d3142887", "locator": "CSVlines2\\u201313901; pc2/pc3 columns",  
"observation": "min=max0 for both columns over 900 rows"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\rtch_e1.py", "sha256":  
"4fe93acda593d564ab204efac217cdf9aa76c256db57d057aa17ef29dc75a1f", "locator": "lines664\\u201313692", "observation": "no  
constant/NaN guard in Spearman permutation function"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\rtch_e1.py", "sha256":  
"37b0763b7e2677ea080ef390588fa65b1486bb5438d0e8e72658641401e888f4", "locator": "lines589\\u201313606", "observation": "same old  
formula"}
```

Interpretation limit: Static logic and stored blanks establish invalid evidentiary status; source execution was not run to reproduce the failure.

Registry action: Exclude undefined-correlation p-values from support, preserve them as diagnostic failures, and require finite/constant-input checks before permutation reporting.

E-LCL-010 - RTCH scalar CV does not validate held-out representation construction

[Certain] RTCH builds median imputation/scaling, PCA, kNN geodesics and diffusion coordinates on the complete analysis sample, then performs 5fold Ridge prediction on precomputed scalars. Faltings_height is also part of intrinsic features and later used as a prediction target. Its reported $cvR^2 \sim 0.99997$ (old) or 0.98505 (new) is not independent prediction of an excluded invariant.

Severity: HIGH. Claims: RANK-002, RTCH-M001, ECC-001. Experiments: RTCH-E1 (noncanonical local experiment), EXP-CTRL-A03.

Provenance: SIMULATION / NEGATIVE / NULL / HISTORICAL MODEL. Assessment: INCONCLUSIVE; leakage/contamination relevant.

Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\rtch_e1.py", "sha256":  
"4fe93acda593d564ab204efac217cfe9aa76c256db57d057aa17ef29dc75a1f", "locator":  
"lines70\u2013313\u201324,569\u20133603,617\u20133649,835,891\u20133902", "observation": "global representation then  
downstream5fold CV"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\results\\prediction_tests.csv", "sha256":  
"6a2c0490e6d8e82a554037067719f620ffd3210f12c2da356d53fc70a53e06fb", "locator": "CSVline7 faltings_height/pc1", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\RTCH_E1\\results\\prediction_tests.csv", "sha256":  
"30b91aed6a6a687b2fe73cdebed209552e6f41f1af1127e1110e9bc7ca26f3e6", "locator": "CSVline7 faltings_height/pc1", "observation": ""}
```

Interpretation limit: Global unsupervised representation is transductive rather than a demonstrated induction to unseen samples. Target inclusion is explicit in the source feature list; measured leakage effect was not estimated.

Registry action: Set Leakage Status POTENTIAL LEAKAGE for rank generalization and CONFIRMED TARGET CONTAMINATION for Faltings-held-out interpretation; rebuild fold-local embeddings and remove target-defining inputs for independent prediction tests.

E-LCL-011 - RTCH stability rows repeat the whole demo sample

[Certain] Nested stability CSV contains 2 identical whole-sample rows, while result.json stability is empty. Current choose_topology_sample returns a copy when $\text{len}(\text{df}) \leq \text{max_points}$; demo350 with $\text{max_points} 350$ therefore repeats the same entire sample, rather than independent subsampling.

Severity: MEDIUM. Claims: RTCH-M001, ECC-001, TOPO-001. Experiments: RTCH-E1 (noncanonical local experiment).

Provenance: SIMULATION / NEGATIVE / NULL. Assessment: INCONCLUSIVE; stability not demonstrated. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\RTCH_E1\\results\\stability_summary.csv", "sha256":  
"4af317de92879957316bd3bbeb4cd4c1755b4b4adf15f437aa2ae37e2d0fb265", "locator": "CSVlines2\u201313", "observation": "identical  
H0/H1/H2 maxima"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\RTCH_E1\\rtch_e1.py", "sha256":  
"4fe93acda593d564ab204efac217cfe9aa76c256db57d057aa17ef29dc75a1f", "locator": "lines301\u20133310,859\u20133875,939\u20133967",  
"observation": "whole-sample copy at \u2264 limit; no stability population of result.json"}
```

Interpretation limit: Available file hashes identify current code, but old manifests omit exact code attribution. CSV existence is not independent resampling/replication.

Registry action: Record CSV stability output separately from empty JSON summary and require genuine perturbation/subsampling before claiming robustness.

E-LCL-012 - Historical L-cosmo SFR script uses global preprocessing and engineered nonarithmetic rank

[Certain] L-cosmo(s).py performs global median imputation, MinMax fit, massmedian, redshiftbinning and groupmeans before train/test splitting. cosmo_rank is a weighted morphology/confidence/redshift feature, not elliptic-curve rank. L_cosmo_s* columns are constructed/plotted but omitted from the model feature list; this script therefore cannot establish incremental predictive value for L_cosmo. Its groupmean Schechter form differs from the canonical mass-ratio sum.

Severity: HIGH. Claims: COSMO-L001, COSMO-L002, COSMO-R001, SFR-A001, CTRL-A002. Experiments: EXP-L001, EXP-SFR-A01, EXP-CTRL-A03, CTRL-COSMO-001.

Provenance: HISTORICAL MODEL / EXPLORATORY / NEGATIVE DIAGNOSTIC. Assessment: INCONCLUSIVE; reconstruction required. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\L-cosmo(s).py", "sha256":  
"408dca936a416be0f25d1dcc705e302c58a72cfbfb5661441c6297872d5df297", "locator": "lines22,35\u201350\u201397\u2013105",  
"observation": "global transformations and no L_cosmo_s* in modelfeatures"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\L-cosmos.py", "sha256":  
"50dfe4ad9b1d04bb8065625460b3e8ca27f08cf824cb2dfd21cf70e843ff02e4", "locator": "lines35\u201369\u20131378", "observation":  
"engineered cosmo_rank before split"}
```

Interpretation limit: Static leakage routes are directly observed, but no experiment estimating their performance effect was run. Script relative input merged_data.csv is absent in that exact folder; candidate parent-folder datasets exist but execution lineage is undocumented.

Registry action: Retain M/H categories and R evidence state; set preprocessing leakage concern, rename/cross-reference cosmo_rank semantically, bind the actual input revision, and conduct fold-local incremental models before promotion.

E-LCL-013 - Random-feature baseline targets stellar mass and named input lacks target

[Certain] model5.py targets logmstar, not SFR. The discovered SDSSDR18_200000.csv has 18 columns and lacks logmstar, so the named input cannot run the provided target expression as-is. The source samples 5 features from all columns without excluding the target. Reconstructing the declared seed 42 feature sampling on the as-found header gives objid,ra,run,r,u; target inclusion is not confirmed for this exact input.

Severity: HIGH. Claims: CTRL-RF001, SFR-A001. Experiments: EXP-CTRL-A01, EXP-CTRL-A03.

Provenance: HISTORICAL MODEL / NEGATIVE DIAGNOSTIC. Assessment: INCONCLUSIVE; reconstruction blocked by dataset schema/version. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\model5.py", "sha256":  
"7d1d9e3d57e26e37ec4e1a9f9578b9ff5d4291f9e68a66eb327a50f638095d44", "locator": "lines12,16\u201326,32,37", "observation":  
"target logmstar; no target exclusion"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\SDSSDR18_200000.csv", "sha256":  
"9dca2008683fde4b0f7a90af5de85ee846a30a47f6f6fae3968c83dea06fa5d2", "locator": "CSVheader line1", "observation": "18columns, no  
logmstar"}
```

Interpretation limit: The audit did not run model training. Another undocumented dataset revision may have included the required target; all-column sampling only establishes potential target-inclusion risk.

Registry action: Classify as historical random-feature stellar-mass baseline/control candidate, preserve absent-target diagnostic, and require exact input/schema plus explicit outcome exclusion and SFR-specific control reconstruction.

E-LCL-014 - Local CSV registries reuse canonical IDs with conflicting meanings

[Certain] Repository registry CSVs use CLAIM-ACSC-001 etc rather than PDF CORE-001 etc. More seriously, experiment CSV maps EXP-CTRL-A01 to provenance control, EXP-CTRL-A02 to leakage control, contrary to attached PDF random-feature/permutation definitions. Dataset entries remain planned with deferred provenance. These namespaces cannot be silently merged by experiment-string identity.

Severity: HIGH. Claims: CORE-001, CTRL-RF001, CTRL-A001, CTRL-A002, DATA-M002. Experiments: EXP-CTRL-A01, EXP-CTRL-A02, EXP-CTRL-A03, DATA-PROV-001.

Provenance: UNVERIFIED / HISTORICAL MODEL. Assessment: INCONCLUSIVE provenance control. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\claim_evidence_v0.2.csv", "sha256": "ef50ff75fe995633934e7a12a635f76d16333ed7d260ddd1fee7930bb9d433f", "locator": "CSVheader andlines2\\u2013", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\experiment_registry_v0.2.csv", "sha256": "b2a8c69710fedfa708d763eebf150cf8b35fe49da8ef8392e9ee0a4625161399", "locator": "CSVlines4\\u2013", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\claim_experiment_crosswalk_v0.2.csv", "sha256": "d8b5bc55187b1ba0f2e6d1e2f0d2a1571c2e92a8e93570de25191892290e34ec", "locator": "CSVlines4\\u2013", "observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\dataset_registry_v0.1.csv", "sha256": "b11f2cf25b903ff1ad32c7d35a6353b3f327d31133f8c816363f26865f78150b", "locator": "CSVlines2\\u2013", "observation": ""}
```

Interpretation limit: Attached Charter/Registries govern this requested update. The repository CSV may represent a distinct historical control schema, not an intent to change canonical PDF identifiers.

Registry action: Keep repository CSV namespace as legacy, preserve original files, and document explicit semantic crosswalk/conflicts before migrating IDs into the canonical update.

E-LCL-015 - Partial upstream download metadata survives for observational sources

[Certain] Download-zone sidecars identify MaNGA HI DR17 v2_0_1 FITS and a SkyServer/CasJobs SDSS CSV export. They provide partial upstream attribution; they do not include the full SDSS SQL, release/query selection, acquisition hash, or complete raw-to-model lineage. The ecdata README identifies the Cremona database and a Zenodo release DOI, but the selected derived arithmetic manifests do not bind to a specific commit/release.

Severity: INFO. Claims: DATA-M002. Experiments: DATA-PROV-001, EXP-DATA-A01.

Provenance: UNVERIFIED / HISTORICAL MODEL. Assessment: Partial data infrastructure. Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\mangaHlall.fits\\uf03aZone.Identifier", "sha256": "de193697f18a34335d7a0c99b8d4fbb3ab730cd3ca76e83a667e2aa3aeb8dfc0", "locator": "ZoneTransfer ReferrerUrl and HostUrl", "observation": "https://data.sdss.org/sas/dr17/manga/Hi/v2_0_1/mangaHlall.fits"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\SDSSDR18_200000.csv\\uf03aZone.Identifier", "sha256": "91ff0b446804f8dfe2aa0e2066df08629b2a4ac12a7f9f6b28825679af038b84", "locator": "ZoneTransfer HostUrl", "observation": "https://skyservice.pha.jhu.edu/CasJobsOutput/CSV/MyTable_pmqr771_0.csv"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\ecdata\\README.md", "sha256": "619466f9a1542c3bfeaa68b592feb903edd205c652cac59d7f21d70444203bb5", "locator": "lines4\\u2013", "observation": "Cremona Database; DOI10.5281/zenodo.161341"}
```

Interpretation limit: URLs and DOI are recorded from local metadata and were not revalidated online. As-found hashes do not establish acquisition-date identity.

Registry action: Preserve upstream identifiers as observed provenance fields; add exact release/query and processing/code bindings before declaring complete lineage.

E-LCL-016 - STAR map visualization uses illustrative synthetic data and preselected Hubble calibration

[Certain] Recovered S.T.A.R.M.A.P. catalog explicitly labels its data illustrative, combines 12 hardcoded example curves with 148 synthetically generated rows, and seeds generation with 0x53544152. The H_eff implementation states calibration so high-z recovers Planck and low-z approaches SH0ES, with hardcoded correction coefficients. Geometry angles are hash-derived from curve labels; cluster summaries are hardcoded. These visualizations do not supply independent observational or arithmetic validation.

Severity: HIGH. Claims: CORE-001, MAP-001, MAP-002, COSMO-E002. Experiments: EXP-MAP-A01, EXP-COSMO-B02, DATA-PROV-001.

Provenance: SIMULATION / HISTORICAL MODEL / EXPLORATORY. Assessment: INCONCLUSIVE; model construction only.

Reproducibility: NOT ATTEMPTED (source execution); audit recomputed file hashes and selected data diagnostics only. Replication: NOT ESTABLISHED.

```
{"path": "/home/kepler/S.T.A.R.M.A.P./src/lib/star/catalog.ts", "readable_audit_copy": "C:/Users/pmqr7/codexl/chatgpt-projects/g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00088.ts", "sha256": "f585130c982be4789c4bc701119b1a1a5fdbc72bd475700b01d48c8c921fbb29", "locator": "lines27\\u201340,43\\u2013387,91\\u2013399", "observation": "Recovered by native WSL read; original source preserved"}
```

```
{
  "path": "/home/kepler/S.T.A.R.M.A.P./src/lib/star/physics.ts",
  "readable_audit_copy": "C:/Users/pmqr7/.codex/.chatgpt-projects/g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00085.ts",
  "sha256": "a9125fddf3570012c8f1a3467562fecf31132fecf301fd3decdf300e59bfe234",
  "locator": "lines27\u2013117\u201313126",
  "observation": "Recovered by native WSL read; original source preserved"
}

{"path": "/home/kepler/S.T.A.R.M.A.P./src/routes/hubble.tsx", "readable_audit_copy": "C:/Users/pmqr7/.codex/.chatgpt-projects/g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00063.tsx", "sha256": "0ac7b166116e1d3ae57fcc57a1e9435a4e5c7b3d26d70dc34349601c8b24b24d", "locator": "lines13\u2013125\u201313149", "observation": "Recovered by native WSL read; original source preserved"}

{"path": "/home/kepler/S.T.A.R.M.A.P./README.md", "readable_audit_copy": "C:/Users/pmqr7/.codex/.chatgpt-projects/g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00005.md", "sha256": "e51bbff49db1e33900f05f5082985a576f18a0de8580e733a78e75680dacc7e3", "locator": "Vision and Implementation Status sections", "observation": "Recovered by native WSL read; original source preserved"}
```

Interpretation limit: Static inspection identifies source data origin and calibration. The 12 hardcoded examples were not revalidated against external catalogs; synthetic invariants were not verified as mutually consistent elliptic curves. The visualization was not executed.

Registry action: Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation.

Targeted Drive documentary findings

E-DRV-001 - K convergence and unstable small-sample values are historical theoretical/simulation reports

[Certain] The Cosmological L-function source describes a log-normal expectation argument and reports $K=0.01176$ for $N=9$ and $K_{\text{sub}}=0.00322$ for $N=5$. These are retained as theoretical/historical and negative/diagnostic provenance; this audit did not rerun the simulations or establish a BSD/cosmological physical correspondence.

Claims: MATH-K001, MATH-K002, MATH-K003, COSMO-L001. Experiments: EXP-K001, EXP-K002.

```
{
  "drive_id": "1LzJvyyL_WSAGDe--ldU8LhOupC3fU8qLG6fzcFbSJSY",
  "title": "Cosmological_L-function+Galaxy_Mass_Distribution",
  "url": "https://docs.google.com/document/d/1LzJvyyL_WSAGDe--ldU8LhOupC3fU8qLG6fzcFbSJSY/edit",
  "locator": "lines 86-91, 116-122",
  "snapshot_text_sha256": "e056655477defaaf2eaa4cec28f4a3d564ec408fd6f333bb1cae353d995383f3",
  "digest_type": "SHA256 of connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest",
  "snapshot_file": "1LzJvyyL_WSAGDe--ldU8LhOupC3fU8qLG6fzcFbSJSY.txt"
}
```

Registry action: Reconstruct assumptions, exact masses/units, median definition, seeds and sample selection. Retain unstable results and separate the expectation argument from physical interpretation.

E-DRV-002 - Drive SFR source contains preprocessing before the split in inspected script variants

[Certain] The SFR Prediction Model Refinement snapshot contains an imputer fit_transform at line 140 before train_test_split at line 161, and other whole-data feature construction variants. This is a static validation-design concern for those variants; execution of that exact variant and the magnitude of any performance effect were not established.

Claims: SFR-A001, SFR-E001, SFR-A003, COSMO-L002, CTRL-A002. Experiments: EXP-CTRL-A03, EXP-SFR-A01, EXP-SFR-A03.

```
{
  "drive_id": "1bzrY4bg5_Vi3RucO7NiIkPtYpI8GptmqlRtP_WF_qaw",
  "title": "SFR_Prediction_Model_Refinement",
  "url": "https://docs.google.com/document/d/1bzrY4bg5_Vi3RucO7NiIkPtYpI8GptmqlRtP_WF_qaw/edit",
  "locator": "lines 117-166, 413-424",
  "snapshot_text_sha256": "d2d83ffc304678df0813d76491dc8442789f0c831f47f6d120029cdcf41ac2a3",
  "digest_type": "SHA256 of connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest",
  "snapshot_file": "1bzrY4bg5_Vi3RucO7NiIkPtYpI8GptmqlRtP_WF_qaw.txt"
}
```

Registry action: Bind a precise executed code/data revision and reconstruct fold-local preprocessing before crediting reported predictive performance.

E-DRV-003 - Five-fold validation document does not itself establish independent validation

[Certain] The 5 Fold Validation document contains preprocessing/derived-statistic code and prose describing cross-validation as robustness. A readable document or a stated five-fold procedure is not a linked independent dataset execution; the inspected material does not satisfy the charter full data-code-parameters-null-results chain by itself.

Claims: SFR-A001, COSMO-L001, COSMO-L002, CTRL-A002. Experiments: EXP-L001, EXP-SFR-A01, EXP-CTRL-A03.

```
{"drive_id": "1qSkMYPbRGytMD4qEB-IlHcBt_V3cXb4nAQs7JtAo96U", "title": "5 Fold Validation", "url":  
"https://docs.google.com/document/d/1qSkMYPbRGytMD4qEB-IlHcBt_V3cXb4nAQs7JtAo96U/edit", "locator": "lines 1-48",  
"snapshot_text_sha256": "5056614627b541bc35432235a1636b9d87afd8e4c9642c3f335c56d62013ee36", "digest_type": "SHA256 of  
connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest", "snapshot_file":  
"1qSkMYPbRGytMD4qEB-IlHcBt_V3cXb4nAQs7JtAo96U.txt"}
```

Registry action: Retain proposal/code provenance; validate an exact runnable variant, training-fold derived quantities, outcome definition and locked independent test.

E-DRV-004 - Drive raw data log mixes code proposals and conversational interpretations

[Certain] The raw data log includes research interpretations, script variants and conversational requests to run future steps. These establish documentary provenance, not authenticated executions. Source directions were treated as source text and were not followed.

Claims: CORE-001, DATA-M002, ECC-001. Experiments: EXP-DATA-A02, EXP-ECC-B01.

```
{"drive_id": "1ZVWuuwAS8zRrEsLIYNNSFNuZRGjUWdHbl4WSxpFVreo", "title": "raw data log", "url":  
"https://docs.google.com/document/d/1ZVWuuwAS8zRrEsLIYNNSFNuZRGjUWdHbl4WSxpFVreo/edit", "locator": "lines 11-15, 48-187,  
2438-2472", "snapshot_text_sha256": "ccfc95874403c889504493591f95f618ba69ec3421e433c6eb2fbd9fee19c27d", "digest_type":  
"SHA256 of connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest", "snapshot_file":  
"1ZVWuuwAS8zRrEsLIYNNSFNuZRGjUWdHbl4WSxpFVreo.txt"}
```

Registry action: Keep proposal, observed output and interpretation separate; require machine run manifests for execution claims.

Registry completeness and source preservation

All 51 claims, 25 canonical experiments and 51 current crosswalk records are accounted for. Ten original crosswalk experiment references lack full definitions in the supplied canonical registry. Their literal IDs remain preserved; possible aliases are explicitly unconfirmed.

All three native Google Docs received the complete audit supplement. Connector readback verified the full supplemental text, original baseline blocks unchanged except shifted indices, original tab identities, one audit-date chip and five native source citations in each. Native Google Docs visual rendering was not verified; the local PDF supplements are separately rendered and inspected.

Code extraction log

Local: 4,365 saved artifacts; 20,016 source occurrences; 4,365 saved artifact hashes verified. 931 Python artifacts parse without execution.

Drive: 3,930 saved artifacts; 7,164 source occurrences; 3,930 saved artifact hashes verified. 1,153 Python artifacts parse without execution.

Combined log: 1,230 direct script, notebook-cell and explicit-fence artifacts; 7,065 inferred document/OCR candidates separated for review. Cross-root identical digests occur for 80 artifacts; both provenance records are retained.

The OneDrive code log retains individual code files, source locators, source/snapshot digests and code digests. Exact duplicates retain all occurrence records; variants remain separate. Explicit source files, notebook cells and fenced code are distinguished from inferred snippets. PDF and inferred snippets can require boundary, indentation or glyph review. Parsing does not establish scientific correctness or executability. No extracted code was executed.

Open extraction gaps and exclusions

local_code_manifest.json: 1718 recorded issues. Complete issue and exclusion lists are in the machine-readable manifests.

drive_code_manifest.json: 0 recorded issues. Complete issue and exclusion lists are in the machine-readable manifests.

157 unique likely-code source images OCR scanned from 314 paths; routine generated charts not treated as code; other/archive images recorded for visual review

PDF extraction candidates may span page boundaries; source review required

Drive audio/video have no accessible code transcript and were not downloaded

OneDrive Research-Analysis+Testing_pt.4.zip is a cloud placeholder whose directory read stalled; metadata only; large locally readable archive containers lack whole-container hashes but selected code member bytes are hashed

Both XLSX datasets and the EPUB were inspected; no unsupported flag remains for those

Third-party dependencies/runtimes/credentials excluded with explicit path records

Two Drive media assets were not transcribed. Blank pages and broken-image placeholders cannot yield missing screenshot code. Runtime, dependency, credential and third-party vendor code is excluded as recorded in the log. The registry update confers no new independent replication or physical validation.